

Qatar-1b

R-0596

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_231204 · created 2026-07-25T01:34:56+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460282.5975 +/- 0.0026 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.128 +/- 0.043
Transit depth (Rp/Rs)^2	1.63 +/- 1.11 [%]
Orbital Inclination (inc)	88.94 +/- 2.84
Airmass coefficient 1 (a1)	0.989 +/- 0.012
Airmass coefficient 2 (a2)	0.006 +/- 0.035
Scatter in the residuals of the lightcurve fit is	1.22 %
Best Comparison Star	#2 - [272, 42]
Optimal Aperture	3.55
Optimal Annulus	11.05
Transit Duration (day)	0.047 +/- 0.0042

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.128 $\pm$ 0.043 vs 0.1463 (deviation 0.4 σ)
Duration fit vs published	0.047 d vs 0.0692 d (deviation 4.92 σ)
Mid-time O-C	6.7 min
Depth SNR	2.8
Residual scatter	1.22 %

Light curve

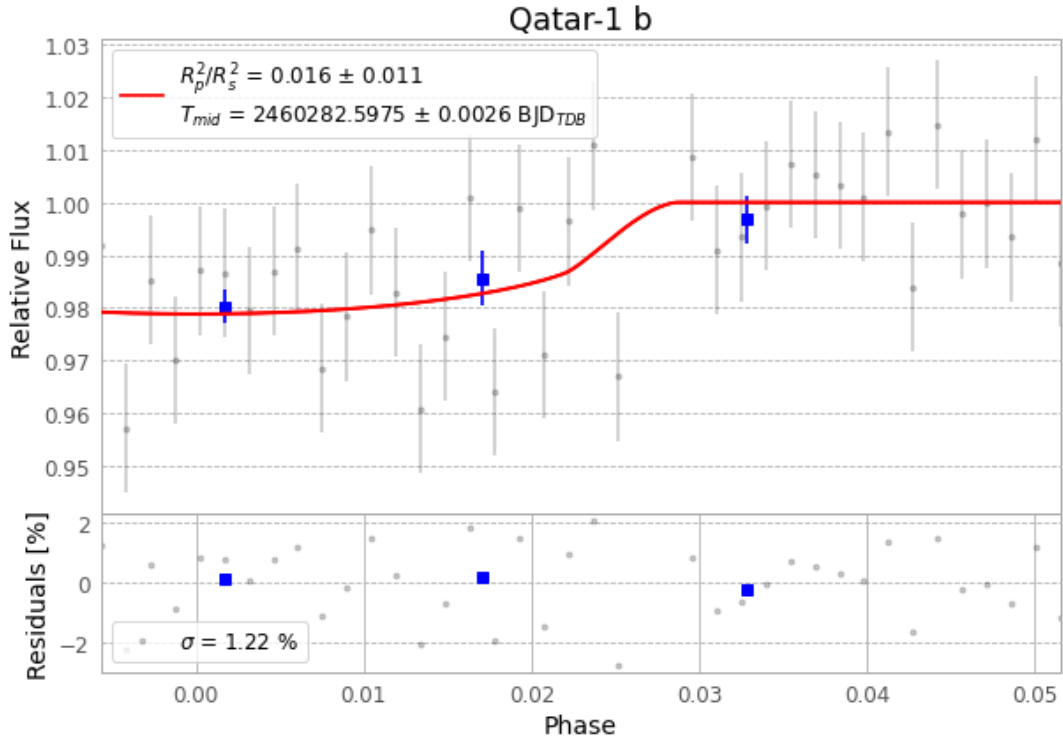


Figure 1 — FinalLightCurve_Qatar-1 b_2023-12-03.png (SHA-256 f4fdf0f740f188e5...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [298, 197], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b557d70065266a9b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

38 FITS frames (25 MB), Qatar-1231204020615.FITS → Qatar-1231204040315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-231204.log (492c357bb1d5...), FinalLightCurve_Qatar-1 b_2023-12-03.png (f4fdf0f740f1...), FinalLightCurve_Qatar-1 b_2023-12-03.csv (d0df85653de1...)

Compute resources

Wall time 95.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 01:34Z.